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United States Patent	12416264
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Desjardins; Michel et al.

Lubrication system for aircraft powerplant and clutch system

Abstract

A system is provided for an aircraft. This aircraft system includes a powertrain and a first lubrication system. The powertrain includes a powerplant, a bladed rotor and a clutch system operatively coupling the powerplant to the bladed rotor. The powerplant is configured to drive rotation of the bladed rotor through the clutch system when the clutch system is engaged. The first lubrication system is fluidly coupled with and is configured to deliver first lubricant to the powerplant and the clutch system.

Inventors: Desjardins; Michel (Saint-Hubert, CA), Durocher; Eric S. (Boucherville, CA), Landry-Drolet; Guillaume (Boucherville, CA)

Applicant: Pratt & Whitney Canada Corp. (Longueuil, CA)

Family ID: 1000008818217

Assignee: Pratt & Whitney Canada Corp. (Longueuil, CA)

Appl. No.: 18/381066

Filed: October 17, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250122841 A1	Apr. 17, 2025

Publication Classification

Int. Cl.: F02C7/36 (20060101); B64D35/08 (20250101); F02C7/06 (20060101); F02C7/14 (20060101)

U.S. Cl.:

CPC F02C7/36 (20130101); F02C7/06 (20130101); F05D2260/4023 (20130101); F05D2260/98 (20130101)

Field of Classification Search

CPC: B64C (29/0033); B64C (27/12); B64C (27/14); B64C (27/28); B64C (27/52); B64C (27/06); B64C (27/00); F01D (25/20); F01D (15/08); F01D (25/183); F01D (15/10); F01D (25/18); F01D (5/02); F01D (9/065); F01D (25/36); F02C (7/06); F02C (7/32); F02C (7/36); B64D (35/00); B64D (27/026); F05D (2260/98); F05D (2260/4023); F05D (2220/32); F05D (2240/60); F04D (29/063)

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Primary Examiner: Zamora Alvarez; Eric J

Attorney, Agent or Firm: Getz Balich LLC

Background/Summary

TECHNICAL FIELD

(1) This disclosure relates generally to an aircraft and, more particularly, to lubricating powertrain components of the aircraft.

BACKGROUND INFORMATION

(2) Various systems and methods are known in the art for delivering lubricant to components of an aircraft. While these known lubricant delivery systems and methods have various benefits, there is still room in the art for improvement.

SUMMARY

(3) According to an aspect of the present disclosure, a system is provided for an aircraft. This aircraft system includes a powertrain and a first lubrication system. The powertrain includes a powerplant, a bladed rotor and a clutch system operatively coupling the powerplant to the bladed rotor. The powerplant is configured to drive rotation of the bladed rotor through the clutch system when the clutch system is engaged. The first lubrication system is fluidly coupled with and is configured to deliver first lubricant to the powerplant and the clutch system.

(4) According to another aspect of the present disclosure, another system is provided for an aircraft. This aircraft system includes a powertrain and a lubrication system. The powertrain includes a powerplant, a bladed rotor and a clutch system operatively coupling the powerplant to the bladed rotor. The powerplant is configured to drive rotation of the bladed rotor through the clutch system when the clutch system is engaged. The lubrication system includes a heat exchanger and a lubricant circuit. The heat exchanger includes an inlet and an outlet. The lubricant circuit extends longitudinally from the outlet to the inlet and through the powerplant and the clutch system.

(5) According to still another aspect of the present disclosure, another system is provided for an aircraft. This aircraft system includes a powertrain, a first lubrication system and a second lubrication system. The powertrain includes a powerplant, a bladed rotor, a geartrain and a clutch system. The powerplant is configured to drive rotation of the bladed rotor through the clutch system and the geartrain. The geartrain operatively couples the bladed rotor to the clutch system. The clutch system operatively couples the geartrain to the powerplant when the clutch system is engaged. The first lubrication system is configured to deliver a first lubricant to the powerplant and the clutch system. The second lubrication system is configured to deliver a second lubricant to the geartrain. The second lubrication system is fluidly independent of the first lubrication system.

(6) The powertrain may also include a second geartrain operatively coupling the powerplant to the clutch system. The first lubrication system may also be configured to deliver the first lubricant to the second geartrain.

(7) The powertrain may also include a power transmission operatively coupling the powerplant to the clutch system. The lubricant circuit may also extend through the power transmission.

(8) The lubricant circuit may extend sequentially from the power transmission, through the powerplant, to the clutch system.

(9) The powerplant may also include a power transmission operatively coupling the powerplant to the clutch system. The first lubrication system may be fluidly coupled to and may also be

configured to deliver the first lubricant to the power transmission.

(10) The power transmission may be configured as or otherwise include a geartrain operatively coupling the powerplant to the clutch system. The first lubrication system may be fluidly coupled to and may also be configured to deliver the first lubricant to the geartrain.

(11) The power transmission may be configured as or otherwise include an accessory geartrain. The first lubrication system may be fluidly coupled to and may also be configured to deliver the first lubricant to the accessory geartrain.

(12) The first lubrication system may also include a heat exchanger and a lubricant circuit. The heat exchanger may include an inlet and an outlet. The lubricant circuit may extend longitudinally through the power transmission, the powerplant and the clutch system from the outlet to the inlet.

(13) The first lubrication system may include a lubricant circuit extending in a downstream direction from the power transmission to the powerplant.

(14) The first lubrication system may include a lubricant circuit extending in a downstream direction from the powerplant to the power transmission.

(15) The first lubrication system may include a lubricant circuit extending in a downstream direction from the power transmission to the clutch system.

(16) The first lubrication system may include a lubricant circuit extending in a downstream direction from the powerplant to the clutch system.

(17) The powerplant may be configured as or otherwise include an internal combustion engine.

(18) The powerplant may be configured as or otherwise include an electric motor.

(19) The system may also include a geartrain and a second lubrication system. The geartrain may operatively couple the clutch system to the bladed rotor. The second lubrication system is fluidly independent of the first lubrication system. The second lubrication system is fluidly coupled with and configured to deliver second lubricant to the geartrain.

(20) The system may also include a geartrain operatively coupling the clutch system to the bladed rotor. The first lubrication system is fluidly independent of the geartrain.

(21) The bladed rotor may be configured as or otherwise include a rotorcraft rotor. The aircraft may be configured as or otherwise include a rotorcraft.

(22) The rotorcraft rotor may be configured as or otherwise include a main rotorcraft rotor.

(23) The present disclosure may include any one or more of the individual features disclosed above and/or below alone or in any combination thereof.

(24) The foregoing features and the operation of the invention will become more apparent in light of the following description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a partial schematic illustration of an aircraft powertrain.

(2) FIG. 2 is a partial schematic illustration of an aircraft system which includes the aircraft powertrain and one or more lubrication systems.

(3) FIG. 3 is a partial schematic illustration of the aircraft system with another lubrication system arrangement.

DETAILED DESCRIPTION

(4) FIG. 1 schematically illustrates a powertrain **10** for an aircraft such as a manned or unmanned rotorcraft; e.g., a helicopter, a drone, etc. The aircraft powertrain **10** includes one or more rotor systems **12** and **14** and a power system **16** for mechanically driving operation of the rotor systems **12** and **14**.

(5) The first rotor system **12** of FIG. 1 includes a first bladed rotor **18** (e.g., an air mover) and a first gear system **20**. The first bladed rotor **18** may be configured as a main rotor for the aircraft (e.g., a

rotorcraft main rotor), and the first gear system **20** may be configured as a main rotor gearbox of the aircraft powertrain **10**.

(6) The first bladed rotor **18** of FIG. **1** is configured to rotate about a first rotor axis **21** (e.g., a vertical axis) to generate aircraft lift and/or aircraft thrust. The first bladed rotor **18** includes a first rotor base **22** (e.g., a hub) and a plurality of first rotor blades **24** arranged circumferentially around, projecting radially out from and connected to the first rotor base **22**.

(7) The first gear system **20** is operatively coupled to the first bladed rotor **18** and its first rotor base **22** through at least one transmission device such as a first rotor shaft **26**; e.g., a main rotor mast. The first gear system **20** of FIG. **1** includes an input **28**, an output **30** and a geartrain **32**; e.g., internal gearing. The first gear system input **28** is coupled to the power system **16**. The first gear system output **30** is coupled to the first bladed rotor **18** through the first rotor shaft **26**. The first gear system geartrain **32** couples the first gear system input **28** to the first gear system output **30**. The first gear system geartrain **32** may be configured such that the first gear system input **28** is operable to rotate at a different (e.g., faster or slower) rotational speed than the first gear system output **30**. The first gear system **20** may thereby be configured as a speed change device between the power system **16** and the first bladed rotor **18**. The first gear system geartrain **32** may also (or alternatively) be configured to change a direction of power transmission through the first gear system **20**. For example, the first gear system output **30** may rotate with the first bladed rotor **18** about the first rotor axis **21**, and the first gear system input **28** may rotate about another axis **34** which is angularly offset from (e.g., perpendicular to) the first rotor axis **21**. The present disclosure, however, is not limited to such an exemplary first gear system arrangement. The first gear system **20** and its first gear system geartrain **32**, for example, may alternatively be configured such that the first gear system input **28** and the first gear system output **30** rotate at a common (the same) rotational speed.

(8) The second rotor system **14** of FIG. **1** includes a second bladed rotor **36** (e.g., an air mover) and a second gear system **38**. The second bladed rotor **36** may be configured as a tail rotor for the aircraft (e.g., a rotorcraft tail rotor), and the second gear system **38** may be configured as a tail rotor gearbox of the aircraft powertrain **10**.

(9) The second bladed rotor **36** of FIG. **1** is configured to rotate about a second rotor axis **40** (e.g., a horizontal axis) to generate lateral thrust to counteract a torque effect associated with rotation of the first bladed rotor **18**. The second bladed rotor **36** includes a second rotor base **42** (e.g., a hub) and a plurality of second rotor blades **44** arranged circumferentially around, projecting radially out from and connected to the second rotor base **42**.

(10) The second gear system **38** is operatively coupled to the second bladed rotor **36** and its second rotor base **42** through at least one transmission device such as a second rotor shaft **46**; e.g., a tail rotor shaft. The second gear system **38** of FIG. **1** includes an input **48**, an output **50** and a geartrain **52**; e.g., internal gearing. The second gear system input **48** is coupled to the power system **16**. The second gear system output **50** is coupled to the second bladed rotor **36** through the second rotor shaft **46**. The second gear system geartrain **52** couples the second gear system input **48** to the second gear system output **50**. The second gear system geartrain **52** may be configured such that the second gear system input **48** is operable to rotate at a different (e.g., faster or slower) rotational speed than the second gear system output **50**. The second gear system **38** may thereby be configured as a speed change device between the power system **16** and the second bladed rotor **36**. The second gear system geartrain **52** may also (or alternatively) be configured to change a direction of power transmission through the second gear system **38**. For example, the second gear system output **50** may rotate with the second bladed rotor **36** about the second rotor axis **40**, and the second gear system input **48** may rotate about another axis **54** which is angularly offset from (e.g., perpendicular to) the second rotor axis **40**. The present disclosure, however, is not limited to such an exemplary second gear system arrangement. The second gear system **38** and its second gear system geartrain **52**, for example, may alternatively be configured such that the second gear system

input **48** and the second gear system output **50** rotate at a common (the same) rotational speed.

Moreover, in other embodiments, the second rotor shaft **46** may be omitted and the second bladed rotor **36** may be connected (e.g., directly) to the second gear system output **50**.

(11) The power system **16** of FIG. **1** includes an aircraft powerplant **56**, a power transmission **58** and a clutch system **60**. The aircraft powerplant **56** may be configured as or otherwise include a gas turbine engine (e.g., a turboshaft engine), a rotary engine (e.g., a Wenkel engine) or any other type of internal combustion engine capable of generating mechanical power for driving rotation of the bladed propulsor rotors **18** and **36**. The aircraft powerplant **56** may alternatively be configured as or otherwise include a hybrid powerplant which includes internal combustion engine and one or more electric motors, or a full electric powerplant which includes one or more electric motors.

(12) The power transmission **58** is configured to operatively couple the aircraft powerplant **56** to the clutch system **60**. The power transmission **58** of FIG. **1**, for example, includes an input **62**, an output **64** and a geartrain **66**; e.g., internal gearing such as an epicyclic gear system or another type of non-epicyclic gear system. The power transmission input **62** is coupled to an output **68** (e.g., an output shaft) from the aircraft powerplant **56**. The power transmission output **64** is coupled to an input **70** into the clutch system **60**. The power geartrain **66** couples the power transmission input **62** to the power transmission output **64**. The power geartrain **66** may be configured such that the power transmission input **62** is operable to rotate at a different (e.g., faster or slower) rotational speed than the power transmission output **64**. This power transmission **58** may thereby be configured as a speed change device between the aircraft powerplant **56** and the clutch system **60**. The power geartrain **66** may also (or alternatively) be configured to change a direction of power transmission through the power transmission **58**. For example, the power transmission output **64** may rotate about an output axis **72**, and the power transmission input **62** may rotate about an input axis **74** which is angularly offset from (e.g., perpendicular to) the output axis **72**. The present disclosure, however, is not limited to such an exemplary power transmission arrangement. The power transmission **58** and its power geartrain **66**, for example, may alternatively be configured such that the power transmission input **62** and the power transmission output **64** rotate about a common (the same) axis or parallel axes.

(13) The clutch system **60** is configured to selectively operatively couple the power transmission **58** to the first rotor system **12** and/or the second rotor system **14**. The clutch system **60** of FIG. **1**, for example, is configured to operatively couple the power transmission output **64** to the first gear system input **28** and/or the second gear system input **48** when a clutch or clutches within the clutch system **60** are engaged. The clutch system **60** of FIG. **1** is configured to decouple the power transmission output **64** from the first gear system input **28** and/or the second gear system input **48** when the clutch or the clutches within the clutch system **60** are disengaged. With this arrangement, the powerplant may be powered up without driving rotation of the first bladed rotor **18** and/or the second bladed rotor **36** when the clutch system **60** is disengaged and/or the first bladed rotor **18** and/or the second bladed rotor **36** may rotate independent of the powerplant when the clutch system **60** is disengaged.

(14) FIG. **2** schematically illustrates a system **76** of the aircraft which includes the aircraft powertrain **10** (see FIG. **1**) and its members **20**, **52**, **56**, **58** and **60** arranged with one or more lubrication systems **78** and **80**. The power system lubrication system **78** is configured to service (e.g., lubricate, cool, etc.) at least (or only) the power system **16**. The rotor system lubrication system **80** is configured to service (e.g., lubricate, cool, etc.) at least (or only) the first rotor system **12**. These lubrication systems **78** and **80**, however, are configured fluidly independent of one another. The power system lubrication system **78**, for example, does not provide lubricant to nor receive lubricant from the rotor system lubrication system **80**. Similarly, the rotor system lubrication system **80** does not provide lubricant to nor receive lubricant from the power system lubrication system **78**. Each of the lubrication system **78**, **80** may thereby circulate its own discrete lubricant, where the lubricants circulated within the lubrication system **78**, **80** may be a common

type of lubricant or different types of lubricant.

(15) The power system lubrication system **78** is fluidly coupled with and configured to deliver its lubricant to the power system members **56**, **58** and **60**. The power system lubrication system **78** of FIG. **2**, for example, includes a power system lubricant circuit **82**, a power system lubricant heat exchanger **84** and a power system lubricant pump **86**. This power system lubrication system **78**, of course, may also include various other fluid system components, which are not shown in FIG. **2** for clarity of illustration. Examples of these other fluid system components include, but are not limited to, one or more filters, one or more sensors, one or more additional pumps, one or more additional heat exchangers, one or more lubricant collection devices (e.g., sumps, gutters, etc.), and the like.

(16) The power system lubricant circuit **82** extends longitudinally (in a downstream direction) through the power system **16** from an outlet **88** from the power system lubricant heat exchanger **84** to an inlet **90** into the power system lubricant heat exchanger **84**. More particularly, the power system lubricant circuit **82** of FIG. **2** extends longitudinally through an upstream region of the power transmission **58** (and its power geartrain **66**), the aircraft powerplant **56**, a downstream region of the power transmission **58** (and its power geartrain **66**) and the clutch system **60** between the outlet **88** and the inlet **90** of the power system lubricant heat exchanger **84**. Here, the upstream region of the power transmission **58** is upstream of the various power system members **56** and **60** along the power system lubricant circuit **82**. The aircraft powerplant **56** is upstream of the various power system members **58** and **60** along the power system lubricant circuit **82**. The downstream region of the power transmission **58** is upstream of the clutch system **60** along the power system lubricant circuit **82**. The present disclosure, however, is not limited to such an exemplary trajectory through the various power system members **56**, **58** and **60**. For example, the aircraft powerplant **56** may alternatively be arranged (e.g., completely) upstream of the other power source members **58** and **60**. In another example, the aircraft powerplant **56** may be arranged in parallel with the power transmission **58**. The power system lubricant heat exchanger **84** is configured to cool the lubricant passing longitudinally therethrough from its inlet **90** to its outlet **88**. The power system lubricant pump **86** is configured to circulate (e.g., pump) the lubricant within the power system lubrication system **78** and its power system lubricant circuit **82**.

(17) By servicing the power system components **56**, **58** and **60** together with the power system lubricant circuit **82**, seal interfaces between the power system components **56**, **58** and **60** may be simplified since the same lubricant is directed those power system components **56**, **58** and **60**. Moreover, in contrast to providing a separate lubricant circuit for the clutch system **60**, arranging the power system lubricant circuit **82** as described above may reduce an overall part count (e.g., tubes, screens, seals, etc.) for the aircraft system **76**. This may in turn reduce aircraft system weight and/or cost.

(18) The rotor system lubrication system **80** is fluidly coupled with and configured to deliver its lubricant to at least (or only) the first gear system **20**. The rotor system lubrication system **80** of FIG. **2**, for example, includes a rotor system lubricant circuit **92**, a rotor system lubricant heat exchanger **94** and a rotor system lubricant pump **96**. This rotor system lubrication system **80**, of course, may also include various other fluid system components, which are not shown in FIG. **2** for clarity of illustration. Examples of these other fluid system components include, but are not limited to, one or more filters, one or more sensors, one or more additional pumps, one or more additional heat exchangers, one or more lubricant collection devices (e.g., sumps, gutters, etc.), and the like.

(19) The rotor system lubricant circuit **92** extends longitudinally (in a downstream direction) through the first rotor system **12** and its first gear system geartrain **32** from an outlet **98** from the rotor system lubricant heat exchanger **94** to an inlet **100** into the rotor system lubricant heat exchanger **94**. The rotor system lubricant heat exchanger **94** is configured to cool the lubricant passing longitudinally therethrough from its inlet **100** to its outlet **98**. The rotor system lubricant pump **96** is configured to circulate (e.g., pump) the lubricant within the rotor system lubrication system **80** and its first gear system **20**.

(20) In some embodiments, referring to FIG. 3, the power transmission 58 may also include an accessory geartrain 102 (e.g., an accessory gearbox) driven by the aircraft powerplant 56, for example through a power takeoff from the power geartrain 66 or a power takeoff from another spool of the aircraft powerplant 56 (e.g., where the powerplant 56 is a gas turbine engine). Alternatively, the accessory geartrain 102 may be integrated with or included as a part of the power geartrain 66; e.g., a gear of the accessory geartrain 102 may be meshed with a gear of the power geartrain 66, etc. The accessory geartrain 102 is configured to distribute mechanical power to the lubricant pump 86 and/or one or more other accessories for the aircraft system 76. The accessory geartrain 102 may also be serviced (e.g., lubricated, cooled, etc.) by the power system lubrication system 78. The power system lubricant circuit 82, for example, may also extend through the accessory geartrain 102, for example between the aircraft powerplant 56 and the clutch system 60.

(21) While the aircraft powertrain 10 is described above with a single first rotor system 12 and a single second rotor system 14, the present disclosure is not limited to such an exemplary arrangement. The aircraft powertrain 10, for example, may alternatively include multiple of the first rotor systems 12. With such an arrangement, the second rotor system 14 may be omitted as torque from the one of the first rotor systems 12 may counteract torque from another one of the first rotor systems 12. Moreover, while the aircraft is generally described above as a rotorcraft, it is contemplated the lubrication systems 78 and 80 may also be utilized for other types of aircraft.

(22) While various embodiments of the present disclosure have been described, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible within the scope of the disclosure. For example, the present disclosure as described herein includes several aspects and embodiments that include particular features. Although these features may be described individually, it is within the scope of the present disclosure that some or all of these features may be combined with any one of the aspects and remain within the scope of the disclosure. Accordingly, the present disclosure is not to be restricted except in light of the attached claims and their equivalents.

Claims

1. A system for an aircraft, comprising: a powertrain including a powerplant, a bladed rotor and a clutch system, the clutch system operatively coupling the powerplant to the bladed rotor, the powerplant configured to drive rotation of the bladed rotor through the clutch system when the clutch system is engaged; a geartrain operatively coupling the clutch system to the bladed rotor; and a first lubrication system fluidly coupled with and configured to deliver a first lubricant to the powerplant and the clutch system, the first lubrication system fluidly independent of the geartrain, wherein the first lubricant is directed through the clutch system.
2. The system of claim 1, wherein the powerplant further includes a power transmission operatively coupling the powerplant to the clutch system; and the first lubrication system is fluidly coupled to the power transmission and further configured to deliver the first lubricant to the power transmission.
3. The system of claim 2, wherein the power transmission comprises a power geartrain operatively coupling the powerplant to the clutch system; and the first lubrication system is fluidly coupled to the power geartrain and further configured to deliver the first lubricant to the power geartrain.
4. The system of claim 2, wherein the power transmission comprises an accessory geartrain; and the first lubrication system is fluidly coupled to the accessory geartrain and further configured to deliver the first lubricant to the accessory geartrain.
5. The system of claim 2, wherein the first lubrication system includes a heat exchanger including an inlet and an outlet; and a lubricant circuit extending longitudinally through the power transmission, the powerplant and the clutch system from the outlet to the inlet.
6. The system of claim 2, wherein the first lubrication system includes a lubricant circuit extending

in a downstream direction from the power transmission to the powerplant.

7. The system of claim 2, wherein the first lubrication system includes a lubricant circuit extending in a downstream direction from the powerplant to the power transmission.
 8. The system of claim 2, wherein the first lubrication system includes a lubricant circuit extending in a downstream direction from the power transmission to the clutch system.
 9. The system of claim 1, wherein the first lubrication system includes a lubricant circuit extending in a downstream direction from the powerplant to the clutch system.
 10. The system of claim 1, wherein the powerplant comprises an internal combustion engine.
 11. The system of claim 1, wherein the powerplant comprises an electric motor.
 12. The system of claim 1, further comprising a second lubrication system fluidly independent of the first lubrication system, the second lubrication system fluidly coupled with and configured to deliver a second lubricant to the geartrain.
 13. The system of claim 1, wherein the bladed rotor comprises a rotorcraft rotor; and the aircraft comprises a rotorcraft.
 14. The system of claim 13, wherein the rotorcraft rotor is a main rotorcraft rotor.
 15. A system for an aircraft, comprising: a powertrain including a powerplant, a bladed rotor and a clutch system, the clutch system operatively coupling the powerplant to the bladed rotor, the powerplant configured to drive rotation of the bladed rotor through the clutch system when the clutch system is engaged; a gear system geartrain operatively coupling the clutch system to the bladed rotor; and a lubrication system including a heat exchanger and a lubricant circuit, the heat exchanger including an inlet and an outlet, the lubricant circuit extending longitudinally from the outlet, through the powerplant and the clutch system, and to the inlet, and the lubrication system fluidly independent of the gear system geartrain.
 16. The system of claim 15, wherein the powertrain further includes a power transmission operatively coupling the powerplant to the clutch system; and the lubricant circuit further extends through the power transmission.
 17. The system of claim 16, wherein the lubricant circuit extends sequentially from the power transmission, through the powerplant, to the clutch system.
 18. The system of claim 15, wherein the lubricant circuit extends in a downstream direction from the outlet, through the powerplant and the clutch system, and to the inlet.
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